Received Order Analysis Engine

Analysis of the King Wen sequence including observations by Terence McKenna

Hexagram Table

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Hexagram reference table
Each hexagram is a stack of 6 lines, either solid (1/yang) or broken (0/yin).
The 'binary' column encodes these 6 lines as bits (bottom line = rightmost bit).
Each hexagram is also composed of two 3-line trigrams: an upper and a lower.
There are 8 possible trigrams (Heaven, Earth, Thunder, Water, Mountain, Wind,
Fire, Lake), giving 8x8 = 64 possible hexagrams.
---
Pos | Hex | Binary | Upper      | Lower      | Name
----|-----|--------|----------------|----------------|----
01  | ䷀  | 111111 | Qian Heaven    | Qian Heaven    | The Creative
02  | ䷁  | 000000 | Kun  Earth     | Kun  Earth     | The Receptive
03  | ䷂  | 010001 | Kan  Water     | Zhen Thunder   | Difficulty at the Beginning
04  | ䷃  | 100010 | Gen  Mountain  | Kan  Water     | Youthful Folly
05  | ䷄  | 010111 | Kan  Water     | Qian Heaven    | Waiting
06  | ䷅  | 111010 | Qian Heaven    | Kan  Water     | Conflict
07  | ䷆  | 000010 | Kun  Earth     | Kan  Water     | The Army
08  | ䷇  | 010000 | Kan  Water     | Kun  Earth     | Holding Together
09  | ䷈  | 110111 | Xun  Wind      | Qian Heaven    | Small Taming
10  | ䷉  | 111011 | Qian Heaven    | Dui  Lake      | Treading
11  | ䷊  | 000111 | Kun  Earth     | Qian Heaven    | Peace
12  | ䷋  | 111000 | Qian Heaven    | Kun  Earth     | Standstill
13  | ䷌  | 111101 | Qian Heaven    | Li   Fire      | Fellowship
14  | ䷍  | 101111 | Li   Fire      | Qian Heaven    | Great Possession
15  | ䷎  | 000100 | Kun  Earth     | Gen  Mountain  | Modesty
16  | ䷏  | 001000 | Zhen Thunder   | Kun  Earth     | Enthusiasm
17  | ䷐  | 011001 | Dui  Lake      | Zhen Thunder   | Following
18  | ䷑  | 100110 | Gen  Mountain  | Xun  Wind      | Work on the Decayed
19  | ䷒  | 000011 | Kun  Earth     | Dui  Lake      | Approach
20  | ䷓  | 110000 | Xun  Wind      | Kun  Earth     | Contemplation
21  | ䷔  | 101001 | Li   Fire      | Zhen Thunder   | Biting Through
22  | ䷕  | 100101 | Gen  Mountain  | Li   Fire      | Grace
23  | ䷖  | 100000 | Gen  Mountain  | Kun  Earth     | Splitting Apart
24  | ䷗  | 000001 | Kun  Earth     | Zhen Thunder   | Return
25  | ䷘  | 111001 | Qian Heaven    | Zhen Thunder   | Innocence
26  | ䷙  | 100111 | Gen  Mountain  | Qian Heaven    | Great Taming
27  | ䷚  | 100001 | Gen  Mountain  | Zhen Thunder   | Nourishment
28  | ䷛  | 011110 | Dui  Lake      | Xun  Wind      | Great Preponderance
29  | ䷜  | 010010 | Kan  Water     | Kan  Water     | The Abysmal
30  | ䷝  | 101101 | Li   Fire      | Li   Fire      | The Clinging
31  | ䷞  | 011100 | Dui  Lake      | Gen  Mountain  | Influence
32  | ䷟  | 001110 | Zhen Thunder   | Xun  Wind      | Duration
33  | ䷠  | 111100 | Qian Heaven    | Gen  Mountain  | Retreat
34  | ䷡  | 001111 | Zhen Thunder   | Qian Heaven    | Great Power
35  | ䷢  | 101000 | Li   Fire      | Kun  Earth     | Progress
36  | ䷣  | 000101 | Kun  Earth     | Li   Fire      | Darkening of the Light
37  | ䷤  | 110101 | Xun  Wind      | Li   Fire      | The Family
38  | ䷥  | 101011 | Li   Fire      | Dui  Lake      | Opposition
39  | ䷦  | 010100 | Kan  Water     | Gen  Mountain  | Obstruction
40  | ䷧  | 001010 | Zhen Thunder   | Kan  Water     | Deliverance
41  | ䷨  | 100011 | Gen  Mountain  | Dui  Lake      | Decrease
42  | ䷩  | 110001 | Xun  Wind      | Zhen Thunder   | Increase
43  | ䷪  | 011111 | Dui  Lake      | Qian Heaven    | Breakthrough
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44 | ䷫ | 111110 | Qian Heaven  | Xun Wind   | Coming to Meet
45 | ䷬ | 011000 | Dui Lake   | Kun Earth  | Gathering Together
46 | ䷭ | 000110 | Kun Earth  | Xun Wind   | Pushing Upward
47 | ䷮ | 011010 | Dui Lake   | Kan Water  | Oppression
48 | ䷯ | 010110 | Kan Water  | Xun Wind   | The Well
49 | ䷰ | 011101 | Dui Lake   | Li  Fire   | Revolution
50 | ䷱ | 101110 | Li  Fire   | Xun Wind   | The Cauldron
51 | ䷲ | 001001 | Zhen Thunder | Zhen Thunder | The Arousing
52 | ䷳ | 100100 | Gen Mountain | Gen Mountain | Keeping Still
53 | ䷴ | 110100 | Xun Wind   | Gen Mountain | Development
54 | ䷵ | 001011 | Zhen Thunder | Dui Lake   | The Marrying Maiden
55 | ䷶ | 001101 | Zhen Thunder | Li  Fire   | Abundance
56 | ䷷ | 101100 | Li  Fire   | Gen Mountain | The Wanderer
57 | ䷸ | 110110 | Xun Wind   | Xun Wind   | The Gentle
58 | ䷹ | 011011 | Dui Lake   | Dui Lake   | The Joyous
59 | ䷺ | 110010 | Xun Wind   | Kan Water  | Dispersion
60 | ䷻ | 010011 | Kan Water  | Dui Lake   | Limitation
61 | ䷼ | 110011 | Xun Wind   | Dui Lake   | Inner Truth
62 | ䷽ | 001100 | Zhen Thunder | Gen Mountain | Small Preponderance
63 | ䷾ | 010101 | Kan Water  | Li  Fire   | After Completion
64 | ䷿ | 101010 | Li  Fire   | Kan Water  | Before Completion
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Reverse/Inverse Pairs

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Reverse vs. Inverse pair analysis
The 64 hexagrams are traditionally grouped into 32 pairs (1-2, 3-4, ..., 63-64).
For each pair, we test two relationships:
 Reverse: flip the hexagram upside down (read the lines top-to-bottom instead of
     bottom-to-top). If the second hexagram upside-down equals the first,
     they are a reverse pair.
 Inverse: toggle every line (solid becomes broken, broken becomes solid).
     If toggling all lines of one hexagram gives the other, they are inverse.
Key finding: EVERY pair in the King Wen sequence is one or the other — none are
unrelated. This perfect pairing structure is unlikely to occur by chance.
---
01 ䷀ Inverse 02 ䷁
03 ䷂ Reverse 04 ䷃
05 ䷄ Reverse 06 ䷅
07 ䷆ Reverse 08 ䷇
09 ䷈ Reverse 10 ䷉
11 ䷊ Reverse 12 ䷋
13 ䷌ Reverse 14 ䷍
15 ䷎ Reverse 16 ䷏
17 ䷐ Reverse 18 ䷑
19 ䷒ Reverse 20 ䷓
21 ䷔ Reverse 22 ䷕
23 ䷖ Reverse 24 ䷗
25 ䷘ Reverse 26 ䷙
27 ䷚ Inverse 28 ䷛
29 ䷜ Inverse 30 ䷝
31 ䷞ Reverse 32 ䷟
33 ䷠ Reverse 34 ䷡
35 ䷢ Reverse 36 ䷣
37 ䷤ Reverse 38 ䷥
39 ䷦ Reverse 40 ䷧
41 ䷨ Reverse 42 ䷩
43 ䷪ Reverse 44 ䷫
45 ䷬ Reverse 46 ䷭
47 ䷮ Reverse 48 ䷯
49 ䷰ Reverse 50 ䷱
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51 ䷳ Reverse 52 ䷲
53 ䷴ Reverse 54 ䷵
55 ䷶ Reverse 56 ䷷
57 ䷸ Reverse 58 ䷹
59 ䷺ Reverse 60 ䷻
61 ䷼ Inverse 62 ䷽
63 ䷾ Reverse 64 ䷿
---
Reverse count pairs: 28/32 (87.5%)
Inverse count pairs: 04/32 (12.5%)
Neither count pairs: 00/32 (0.0%)
```

Difference Wave

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First order of difference (the 'wave')
For each consecutive pair of hexagrams, count how many of the 6 lines change.
This produces a sequence of values from 0 (identical) to 6 (every line flipped).
The resulting 'wave' is central to analysis of the King Wen sequence.
Key observation (McKenna, The Invisible Landscape, 1975): the value 5 never appears — no consecutive hexag
differ by exactly 5 lines. The value 0 also never appears (no duplicates).
---
01 ䷀ to 02 ䷁ difference 6 ******
02 ䷁ to 03 ䷂ difference 2 **
03 ䷂ to 04 ䷃ difference 4 ****
04 ䷃ to 05 ䷄ difference 4 ****
05 ䷄ to 06 ䷅ difference 4 ****
06 ䷅ to 07 ䷆ difference 3 ***
07 ䷆ to 08 ䷇ difference 2 **
08 ䷇ to 09 ䷈ difference 4 ****
09 ䷈ to 10 ䷉ difference 2 **
10 ䷉ to 11 ䷊ difference 4 ****
11 ䷊ to 12 ䷋ difference 6 ******
12 ䷋ to 13 ䷌ difference 2 **
13 ䷌ to 14 ䷍ difference 2 **
14 ䷍ to 15 ䷎ difference 4 ****
15 ䷎ to 16 ䷏ difference 2 **
16 ䷏ to 17 ䷐ difference 2 **
17 ䷐ to 18 ䷑ difference 6 ******
18 ䷑ to 19 ䷒ difference 3 ***
19 ䷒ to 20 ䷓ difference 4 ****
20 ䷓ to 21 ䷔ difference 3 ***
21 ䷔ to 22 ䷕ difference 2 **
22 ䷕ to 23 ䷖ difference 2 **
23 ䷖ to 24 ䷗ difference 2 **
24 ䷗ to 25 ䷘ difference 3 ***
25 ䷘ to 26 ䷙ difference 4 ****
26 ䷙ to 27 ䷚ difference 2 **
27 ䷚ to 28 ䷛ difference 6 ******
28 ䷛ to 29 ䷜ difference 2 **
29 ䷜ to 30 ䷝ difference 6 ******
30 ䷝ to 31 ䷞ difference 3 ***
31 ䷞ to 32 ䷟ difference 2 **
32 ䷟ to 33 ䷠ difference 3 ***
33 ䷠ to 34 ䷡ difference 4 ****
34 ䷡ to 35 ䷢ difference 4 ****
35 ䷢ to 36 ䷣ difference 4 ****
36 ䷣ to 37 ䷤ difference 2 **
37 ䷤ to 38 ䷥ difference 4 ****
38 ䷥ to 39 ䷦ difference 6 ******
39 ䷦ to 40 ䷧ difference 4 ****
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40 ䷿ to 41 ䷾ difference 3 ***
41 ䷾ to 42 ䷽ difference 2 **
42 ䷽ to 43 ䷼ difference 4 ****
43 ䷼ to 44 ䷻ difference 2 **
44 ䷻ to 45 ䷺ difference 3 ***
45 ䷺ to 46 ䷹ difference 4 ****
46 ䷹ to 47 ䷸ difference 3 ***
47 ䷸ to 48 ䷷ difference 2 **
48 ䷷ to 49 ䷶ difference 3 ***
49 ䷶ to 50 ䷵ difference 4 ****
50 ䷵ to 51 ䷴ difference 4 ****
51 ䷴ to 52 ䷳ difference 4 ****
52 ䷳ to 53 ䷲ difference 1 *
53 ䷲ to 54 ䷱ difference 6 ******
54 ䷱ to 55 ䷰ difference 2 **
55 ䷰ to 56 ䷯ difference 2 **
56 ䷯ to 57 ䷮ difference 3 ***
57 ䷮ to 58 ䷭ difference 4 ****
58 ䷭ to 59 ䷬ difference 3 ***
59 ䷬ to 60 ䷫ difference 2 **
60 ䷫ to 61 ䷪ difference 1 *
61 ䷪ to 62 ䷩ difference 6 ******
62 ䷩ to 63 ䷨ difference 3 ***
63 ䷨ to 64 ䷀ difference 6 ******
---
0 line changes total 0
1 line changes total 2
2 line changes total 20
3 line changes total 13
4 line changes total 19
5 line changes total 0
6 line changes total 9

Spark line: ▅▁▃▃▁▁▅▁▁▅▁▅▁▁▁▁▅▁▅▅▁▁▅▅▅▁▅▅▁▁▅▁▁▅▁▁▅▅▅▁▅▁▁▅▁▁▅▅
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Bar Chart

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Difference wave bar chart
A visual representation of the first-order difference wave. Each column shows
how many lines changed in that transition. The horizontal axis is the position
in the King Wen sequence (1-63), the vertical axis is the number of lines
changed (1-6). This makes the wave pattern easier to see than a list of numbers.
---
 6 |#    #  #    ##   #      #   ##|
 5 |#    #  #    ##   #      #   ##|
 4 |# ### # ## # ##  ### # ### ### # # ### # # ##|
 3 |# #### # ## # ####  ## # ## #### #### # ### #### # ### ###|
 2 |##################################################### ####### ###|
 1 |#############################################################|
  +-------------------------------------------------------------+
       1    2    3    4    5    6
       0    0    0    0    0    0
```

Trigram Analysis

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Trigram analysis
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```
Each hexagram is built from two trigrams (3-line figures): an upper and a lower.
There are 8 possible trigrams. This section counts how often each trigram appears
in each position, checks how often the upper or lower trigram changes between
consecutive hexagrams, and shows the full 8x8 transition matrix (which trigram
follows which). A uniform distribution (8 of each) confirms that all 64 possible
upper/lower combinations are used exactly once.
---
Trigram       Upper Lower Total
Kun Earth       8   8   16
Zhen Thunder      8   8   16
Kan Water       8   8   16
Dui Lake       8   8   16
Gen Mountain      8   8   16
Li  Fire       8   8   16
Xun Wind       8   8   16
Qian Heaven       8   8   16

--- Consecutive trigram transitions ---
Upper trigram changes: 59/63 transitions (93.7%)
Lower trigram changes: 58/63 transitions (92.1%)

--- Upper trigram transition matrix ---
Rows = from, Columns = to
     Kun  Zhen Kan  Dui  Gen  Li   Xun  Qian
Kun    0   1   2   1   0   0   2   2
Zhen    0   1   1   1   2   2   0   1
Kan    0   1   0   1   1   2   2   1
Dui    1   1   2   0   1   1   1   1
Gen    2   0   1   1   2   0   2   0
Li    2   1   1   1   1   0   1   0
Xun    0   2   1   2   0   2   0   1
Qian    3   1   0   1   1   1   0   1

--- Lower trigram transition matrix ---
Rows = from, Columns = to
     Kun  Zhen Kan  Dui  Gen  Li   Xun  Qian
Kun    0   4   0   0   0   2   1   1
Zhen    0   1   1   0   1   1   2   2
Kan    1   0   1   2   0   1   1   1
Dui    1   1   1   1   2   1   0   1
Gen    1   0   1   1   1   1   2   1
Li    1   0   1   1   2   1   1   1
Xun    1   1   2   2   1   1   0   0
Qian    3   1   1   1   1   0   1   0
```

Nuclear Hexagrams

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Nuclear hexagram analysis
Every hexagram contains a hidden 'nuclear' hexagram inside it, formed by
extracting the 4 middle lines (lines 2-3-4-5). Lines 2-3-4 form the lower
nuclear trigram, and lines 3-4-5 form the upper nuclear trigram (sharing lines
3 and 4). This creates a chain: each hexagram points to its nuclear hexagram,
which points to its own nuclear hexagram, and so on until the chain loops.
These chains reveal deep structural relationships between hexagrams that are
not obvious from the surface arrangement.
---
01 ䷀ nuclear: 01 ䷀  chain: 01 ䷀ -> 01 ䷀
02 ䷁ nuclear: 02 ䷁  chain: 02 ䷁ -> 02 ䷁
03 ䷂ nuclear: 23 ䷊  chain: 03 ䷂ -> 23 ䷊ -> 02 ䷁ -> 02 ䷁
04 ䷃ nuclear: 24 ䷋  chain: 04 ䷃ -> 24 ䷋ -> 02 ䷁ -> 02 ䷁
```

```
05 ䷅ nuclear: 38 ䷨ chain: 05 ䷅ -> 38 ䷨ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
06 ䷆ nuclear: 37 ䷩ chain: 06 ䷆ -> 37 ䷩ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
07 ䷗ nuclear: 24 ䷒ chain: 07 ䷗ -> 24 ䷒ -> 02 ䷁ -> 02 ䷁
08 ䷘ nuclear: 23 ䷖ chain: 08 ䷘ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
09 ䷅ nuclear: 38 ䷨ chain: 09 ䷅ -> 38 ䷨ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
10 ䷆ nuclear: 37 ䷩ chain: 10 ䷆ -> 37 ䷩ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
11 ䷊ nuclear: 54 ䷵ chain: 11 ䷊ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
12 ䷚ nuclear: 53 ䷴ chain: 12 ䷚ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
13 ䷠ nuclear: 44 ䷫ chain: 13 ䷠ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
14 ䷡ nuclear: 43 ䷪ chain: 14 ䷡ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
15 ䷌ nuclear: 40 ䷧ chain: 15 ䷌ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
16 ䷏ nuclear: 39 ䷨ chain: 16 ䷏ -> 39 ䷨ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
17 ䷐ nuclear: 53 ䷴ chain: 17 ䷐ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
18 ䷑ nuclear: 54 ䷵ chain: 18 ䷑ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
19 ䷗ nuclear: 24 ䷒ chain: 19 ䷗ -> 24 ䷒ -> 02 ䷁ -> 02 ䷁
20 ䷘ nuclear: 23 ䷖ chain: 20 ䷘ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
21 ䷔ nuclear: 39 ䷨ chain: 21 ䷔ -> 39 ䷨ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
22 ䷕ nuclear: 40 ䷧ chain: 22 ䷕ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
23 ䷚ nuclear: 02 ䷁ chain: 23 ䷚ -> 02 ䷁ -> 02 ䷁
24 ䷒ nuclear: 02 ䷁ chain: 24 ䷒ -> 02 ䷁ -> 02 ䷁
25 ䷙ nuclear: 53 ䷴ chain: 25 ䷙ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
26 ䷛ nuclear: 54 ䷵ chain: 26 ䷛ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
27 ䷜ nuclear: 02 ䷁ chain: 27 ䷜ -> 02 ䷁ -> 02 ䷁
28 ䷝ nuclear: 01 ䷀ chain: 28 ䷝ -> 01 ䷀ -> 01 ䷀
29 ䷞ nuclear: 27 ䷚ chain: 29 ䷞ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
30 ䷟ nuclear: 28 ䷡ chain: 30 ䷟ -> 28 ䷡ -> 01 ䷀ -> 01 ䷀
31 ䷠ nuclear: 44 ䷫ chain: 31 ䷠ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
32 ䷡ nuclear: 43 ䷪ chain: 32 ䷡ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
33 ䷢ nuclear: 44 ䷫ chain: 33 ䷢ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
34 ䷣ nuclear: 43 ䷪ chain: 34 ䷣ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
35 ䷤ nuclear: 39 ䷨ chain: 35 ䷤ -> 39 ䷨ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
36 ䷥ nuclear: 40 ䷧ chain: 36 ䷥ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
37 ䷦ nuclear: 64 ䷿ chain: 37 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
38 ䷧ nuclear: 63 ䷾ chain: 38 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
39 ䷨ nuclear: 64 ䷿ chain: 39 ䷨ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
40 ䷩ nuclear: 63 ䷾ chain: 40 ䷩ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
41 ䷪ nuclear: 24 ䷒ chain: 41 ䷪ -> 24 ䷒ -> 02 ䷁ -> 02 ䷁
42 ䷫ nuclear: 23 ䷖ chain: 42 ䷫ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
43 ䷬ nuclear: 01 ䷀ chain: 43 ䷬ -> 01 ䷀ -> 01 ䷀
44 ䷭ nuclear: 01 ䷀ chain: 44 ䷭ -> 01 ䷀ -> 01 ䷀
45 ䷮ nuclear: 53 ䷴ chain: 45 ䷮ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
46 ䷯ nuclear: 54 ䷵ chain: 46 ䷯ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
47 ䷰ nuclear: 37 ䷩ chain: 47 ䷰ -> 37 ䷩ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
48 ䷱ nuclear: 38 ䷨ chain: 48 ䷱ -> 38 ䷨ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
49 ䷲ nuclear: 44 ䷫ chain: 49 ䷲ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
50 ䷳ nuclear: 43 ䷪ chain: 50 ䷳ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
51 ䷴ nuclear: 39 ䷨ chain: 51 ䷴ -> 39 ䷨ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
52 ䷵ nuclear: 40 ䷧ chain: 52 ䷵ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
53 ䷶ nuclear: 64 ䷿ chain: 53 ䷶ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
54 ䷷ nuclear: 63 ䷾ chain: 54 ䷷ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
55 ䷸ nuclear: 28 ䷝ chain: 55 ䷸ -> 28 ䷝ -> 01 ䷀ -> 01 ䷀
56 ䷹ nuclear: 28 ䷝ chain: 56 ䷹ -> 28 ䷝ -> 01 ䷀ -> 01 ䷀
57 ䷺ nuclear: 38 ䷨ chain: 57 ䷺ -> 38 ䷨ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
58 ䷻ nuclear: 37 ䷩ chain: 58 ䷻ -> 37 ䷩ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
59 ䷼ nuclear: 27 ䷚ chain: 59 ䷼ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
60 ䷽ nuclear: 27 ䷚ chain: 60 ䷽ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
61 ䷾ nuclear: 27 ䷚ chain: 61 ䷾ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
62 ䷿ nuclear: 28 ䷝ chain: 62 ䷿ -> 28 ䷝ -> 01 ䷀ -> 01 ䷀
63 ䷾ nuclear: 64 ䷿ chain: 63 ䷾ -> 64 ䷿ -> 63 ䷾
64 ䷿ nuclear: 63 ䷾ chain: 64 ䷿ -> 63 ䷾ -> 64 ䷿

--- Nuclear hexagram frequency ---
How often each hexagram appears as another's nuclear hexagram
01 ䷀ The Creative          appears 4 times
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02 ䷁ The Receptive          appears 4 times
23 ䷖ Splitting Apart         appears 4 times
24 ䷗ Return              appears 4 times
27 ䷚ Nourishment           appears 4 times
28 ䷛ Great Preponderance       appears 4 times
37 ䷤ The Family           appears 4 times
38 ䷥ Opposition           appears 4 times
39 ䷦ Obstruction          appears 4 times
40 ䷧ Deliverance          appears 4 times
43 ䷪ Breakthrough          appears 4 times
44 ䷫ Coming to Meet         appears 4 times
53 ䷴ Development          appears 4 times
54 ䷵ The Marrying Maiden       appears 4 times
63 ䷾ After Completion        appears 4 times
64 ䷿ Before Completion        appears 4 times
```

Line Change Analysis

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Line change positional analysis
Each hexagram has 6 lines (positions 1-6, bottom to top). When consecutive
hexagrams differ, which specific lines are changing? If the King Wen sequence
were random, each line position would change about 50% of the time. Significant
deviation from 50% would suggest the ordering favors changing certain positions.
The co-change matrix shows which pairs of lines tend to flip together in the
same transition — revealing correlated line movements.
---
Line 1: 37/63 ( 58.7%) ####################################
Line 2: 32/63 ( 50.8%) ################################
Line 3: 35/63 ( 55.6%) ###################################
Line 4: 34/63 ( 54.0%) ##################################
Line 5: 35/63 ( 55.6%) ###################################
Line 6: 38/63 ( 60.3%) #######################################

--- Line co-change frequency ---
How often pairs of lines change together in the same transition
   L1  L2  L3  L4  L5  L6
L1  .  18  20  19  19  30
L2  .   .  20  17  23  20
L3  .   .   .  25  16  21
L4  .   .   .   .  20  20
L5  .   .   .   .   .  20
L6  .   .   .   .   .   .
```

Complement Distance

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Complement distance analysis
Every hexagram has an 'opposite' (complement) formed by toggling all 6 lines
(solid becomes broken, broken becomes solid). For example, The Creative (all
solid) and The Receptive (all broken) are complements. This section finds where
each hexagram's complement sits in the King Wen sequence and how far apart they
are. If complements tend to be near each other, it suggests the sequence was
deliberately organized around opposition; if far apart, they may serve as
bookends for larger structural sections.
---
01 ䷀ <-> 02 ䷁ distance:  1 The Creative <-> The Receptive
02 ䷁ <-> 01 ䷀ distance:  1 The Receptive <-> The Creative
03 ䷂ <-> 50 ䷱ distance: 47 Difficulty at the Beginning <-> The Cauldron
```

```
04 ䷃ <-> 49 ䷰ distance: 45 Youthful Folly <-> Revolution
05 ䷄ <-> 35 ䷢ distance: 30 Waiting <-> Progress
06 ䷅ <-> 36 ䷣ distance: 30 Conflict <-> Darkening of the Light
07 ䷆ <-> 13 ䷌ distance: 6 The Army <-> Fellowship
08 ䷇ <-> 14 ䷍ distance: 6 Holding Together <-> Great Possession
09 ䷈ <-> 16 ䷏ distance: 7 Small Taming <-> Enthusiasm
10 ䷉ <-> 15 ䷎ distance: 5 Treading <-> Modesty
11 ䷊ <-> 12 ䷋ distance: 1 Peace <-> Standstill
12 ䷋ <-> 11 ䷊ distance: 1 Standstill <-> Peace
13 ䷌ <-> 07 ䷆ distance: 6 Fellowship <-> The Army
14 ䷍ <-> 08 ䷇ distance: 6 Great Possession <-> Holding Together
15 ䷎ <-> 10 ䷉ distance: 5 Modesty <-> Treading
16 ䷏ <-> 09 ䷈ distance: 7 Enthusiasm <-> Small Taming
17 ䷐ <-> 18 ䷑ distance: 1 Following <-> Work on the Decayed
18 ䷑ <-> 17 ䷐ distance: 1 Work on the Decayed <-> Following
19 ䷒ <-> 33 ䷠ distance: 14 Approach <-> Retreat
20 ䷓ <-> 34 ䷡ distance: 14 Contemplation <-> Great Power
21 ䷔ <-> 48 ䷯ distance: 27 Biting Through <-> The Well
22 ䷕ <-> 47 ䷮ distance: 25 Grace <-> Oppression
23 ䷖ <-> 43 ䷪ distance: 20 Splitting Apart <-> Breakthrough
24 ䷗ <-> 44 ䷫ distance: 20 Return <-> Coming to Meet
25 ䷘ <-> 46 ䷭ distance: 21 Innocence <-> Pushing Upward
26 ䷙ <-> 45 ䷬ distance: 19 Great Taming <-> Gathering Together
27 ䷚ <-> 28 ䷛ distance: 1 Nourishment <-> Great Preponderance
28 ䷛ <-> 27 ䷚ distance: 1 Great Preponderance <-> Nourishment
29 ䷜ <-> 30 ䷝ distance: 1 The Abysmal <-> The Clinging
30 ䷝ <-> 29 ䷜ distance: 1 The Clinging <-> The Abysmal
31 ䷞ <-> 41 ䷨ distance: 10 Influence <-> Decrease
32 ䷟ <-> 42 ䷩ distance: 10 Duration <-> Increase
33 ䷠ <-> 19 ䷒ distance: 14 Retreat <-> Approach
34 ䷡ <-> 20 ䷓ distance: 14 Great Power <-> Contemplation
35 ䷢ <-> 05 ䷄ distance: 30 Progress <-> Waiting
36 ䷣ <-> 06 ䷅ distance: 30 Darkening of the Light <-> Conflict
37 ䷤ <-> 40 ䷧ distance: 3 The Family <-> Deliverance
38 ䷥ <-> 39 ䷦ distance: 1 Opposition <-> Obstruction
39 ䷦ <-> 38 ䷥ distance: 1 Obstruction <-> Opposition
40 ䷧ <-> 37 ䷤ distance: 3 Deliverance <-> The Family
41 ䷨ <-> 31 ䷞ distance: 10 Decrease <-> Influence
42 ䷩ <-> 32 ䷟ distance: 10 Increase <-> Duration
43 ䷪ <-> 23 ䷖ distance: 20 Breakthrough <-> Splitting Apart
44 ䷫ <-> 24 ䷗ distance: 20 Coming to Meet <-> Return
45 ䷬ <-> 26 ䷙ distance: 19 Gathering Together <-> Great Taming
46 ䷭ <-> 25 ䷘ distance: 21 Pushing Upward <-> Innocence
47 ䷮ <-> 22 ䷕ distance: 25 Oppression <-> Grace
48 ䷯ <-> 21 ䷔ distance: 27 The Well <-> Biting Through
49 ䷰ <-> 04 ䷃ distance: 45 Revolution <-> Youthful Folly
50 ䷱ <-> 03 ䷂ distance: 47 The Cauldron <-> Difficulty at the Beginning
51 ䷲ <-> 57 ䷸ distance: 6 The Arousing <-> The Gentle
52 ䷳ <-> 58 ䷹ distance: 6 Keeping Still <-> The Joyous
53 ䷴ <-> 54 ䷵ distance: 1 Development <-> The Marrying Maiden
54 ䷵ <-> 53 ䷴ distance: 1 The Marrying Maiden <-> Development
55 ䷶ <-> 59 ䷺ distance: 4 Abundance <-> Dispersion
56 ䷷ <-> 60 ䷻ distance: 4 The Wanderer <-> Limitation
57 ䷸ <-> 51 ䷲ distance: 6 The Gentle <-> The Arousing
58 ䷹ <-> 52 ䷳ distance: 6 The Joyous <-> Keeping Still
59 ䷺ <-> 55 ䷶ distance: 4 Dispersion <-> Abundance
60 ䷻ <-> 56 ䷷ distance: 4 Limitation <-> The Wanderer
61 ䷼ <-> 62 ䷽ distance: 1 Inner Truth <-> Small Preponderance
62 ䷽ <-> 61 ䷼ distance: 1 Small Preponderance <-> Inner Truth
63 ䷾ <-> 64 ䷿ distance: 1 After Completion <-> Before Completion
64 ䷿ <-> 63 ䷾ distance: 1 Before Completion <-> After Completion

--- Complement distance statistics ---
Mean distance: 12.1
```

```
Median distance: 6
Min distance:  1 (adjacent = pair is an inverse pair)
Max distance:  47
```

Palindromes

```
---
Palindrome analysis of the difference wave
A palindrome reads the same forwards and backwards (like 2,4,6,4,2). Finding
palindromic runs in the difference wave suggests the King Wen sequence contains
deliberate mirror symmetry — sections where the pattern of change rises and falls
in a balanced way. Longer palindromes are less likely to occur by chance and more
likely to reflect intentional design.
---
Found 27 palindromic subsequences (showing longest 20)

Length 7 at positions 11-17: [6,2,2,4,2,2,6] ▙▂▃▂▟
Length 7 at positions 19-25: [4,3,2,2,2,3,4] ▄▂▁▁▁▂▄
Length 7 at positions 39-45: [4,3,2,4,2,3,4] ▄▂▁▃▁▂▄
Length 5 at positions 12-16: [2,2,4,2,2] ▂▁▃▁▂
Length 5 at positions 20-24: [3,2,2,2,3] ▃▁▁▁▃
Length 5 at positions 40-44: [3,2,4,2,3] ▃▁▃▁▃
Length 5 at positions 43-47: [2,3,4,3,2] ▂▃▃▃▂
Length 5 at positions 45-49: [4,3,2,3,4] ▄▂▁▂▄
Length 5 at positions 55-59: [2,3,4,3,2] ▂▃▃▃▂
Length 3 at positions 03-05: [4,4,4] ▄▄▄
Length 3 at positions 07-09: [2,4,2] ▂▄▂
Length 3 at positions 08-10: [4,2,4] ▄▂▄
Length 3 at positions 13-15: [2,4,2] ▂▄▂
Length 3 at positions 18-20: [3,4,3] ▃▄▃
Length 3 at positions 21-23: [2,2,2] ▂▂▂
Length 3 at positions 26-28: [2,6,2] ▂▆▂
Length 3 at positions 27-29: [6,2,6] ▆▂▆
Length 3 at positions 30-32: [3,2,3] ▃▂▃
Length 3 at positions 33-35: [4,4,4] ▄▄▄
Length 3 at positions 35-37: [4,2,4] ▄▂▄

--- Palindrome count by length ---
Length 7: 3 found
Length 5: 6 found
Length 3: 18 found
```

Canon Comparison

```
---
Upper Canon (1-30) vs. Lower Canon (31-64) comparison
The I Ching is traditionally divided into two books: the Upper Canon (hexagrams
1-30) and Lower Canon (hexagrams 31-64). This division is ancient and may reflect
different authorship, time periods, or thematic groupings. This section compares
the mathematical properties of each half: do they have similar difference wave
patterns? Similar yin/yang balance? Does the 'no 5-line transition' property hold
in each half independently, or only in the whole?
---
Metric            Upper (1-30) Lower (31-64)
------------------------------ ------------ --------------
Hexagrams                30       34
Transitions               29       33
Mean line changes          3.38      3.33
Yang lines (total)          86      106
```

```
Yin lines (total)            94       98

Difference distribution:    Upper (1-30) Lower (31-64)
0 line changes               0        0
1 line changes               0        2
2 line changes              12        8
3 line changes               4        8
4 line changes               8       11
5 line changes               0        0
6 line changes               5        4

Upper Canon wave: ▌▁▁▁
```

```
26      0.1112    [      : |# :     ]
27     -0.1109    [      : #| :     ]
28      0.1649    [      : | #:     ]
29      0.0093    [      : | :     ]
30     -0.0969    [      : #| :     ]
31     -0.0480    [      : | :     ]

Significant lags (outside noise band): 0/31
No significant autocorrelation detected — consistent with no hidden periodicity,
but note that N=63 provides limited statistical power to detect weak periodicity.
```

Spectral Analysis

```
---
Spectral analysis (Discrete Fourier Transform)
The DFT decomposes the difference wave into frequency components, like splitting
white light into a rainbow. Each frequency represents a periodic pattern in the
wave. A strong peak at frequency F means the wave has a repeating cycle every
63/F transitions. If the King Wen ordering has hidden periodic structure, the
spectrum will show clear peaks rather than flat noise.
Caveat: with only 63 samples, each of the 31 frequency bins has wide
uncertainty. A flat spectrum may indicate no periodicity, or insufficient data.
---
Number of samples: 63
Mean (DC component): 3.349
White noise floor: 0.1741 (magnitudes near this are not significant)

Freq  Period Magnitude Sig? Spectrum
----  ------ --------- ---- --------
  1   63.0   0.0224    ##
  2   31.5   0.1823    #################
  3   21.0   0.0145    #
  4   15.8   0.1359    ############
  5   12.6   0.2021    ##################
  6   10.5   0.1157    ##########
  7    9.0   0.2399    #####################
  8    7.9   0.1092    ##########
  9    7.0   0.0553    #####
 10    6.3   0.1630    ###############
 11    5.7   0.2371    #######################
 12    5.2   0.0473    ####
 13    4.8   0.2326    #####################
 14    4.5   0.0331    ###
 15    4.2   0.1145    ##########
 16    3.9   0.2204    ####################
 17    3.7   0.1451    #############
 18    3.5   0.1975    ##################
 19    3.3   0.2197    ####################
 20    3.1   0.2166    ####################
 21    3.0   0.1611    ###############
 22    2.9   0.0222    ##
 23    2.7   0.1210    ###########
 24    2.6   0.4266  * ########################################
 25    2.5   0.0306    ##
 26    2.4   0.2297    #####################
 27    2.3   0.0795    #######
 28    2.2   0.0639    #####
 29    2.2   0.2114    ###################
 30    2.1   0.2129    ###################
 31    2.0   0.2033    ###################
```

```
Frequencies above 2x noise floor: 1/31
```

Markov Chain

```
---
Markov chain analysis
If we treat each difference value as a 'state', we can ask: does the current
difference predict the next one? In a random sequence, knowing that the last
transition changed 2 lines tells you nothing about the next transition. But if
certain patterns like '6 is always followed by 2' emerge, it reveals hidden
rules in the King Wen ordering. The matrix shows the probability of each
transition between difference values.
---
Transition probability matrix (row = from, column = to)
    -> 1  -> 2  -> 3  -> 4  -> 6
 1  0.00  0.00  0.00  0.00  1.00  (n=2)
 2  0.05  0.25  0.25  0.30  0.15  (n=20)
 3  0.00  0.46  0.00  0.46  0.08  (n=13)
 4  0.05  0.26  0.26  0.32  0.11  (n=19)
 6  0.00  0.50  0.38  0.12  0.00  (n=8)

--- Notable transition patterns ---
(Caveat: rows with small n are unreliable — patterns based on <5 observations
are anecdotes, not statistically meaningful.)
 3 -> 2: 46% (6/13 times)
 3 -> 4: 46% (6/13 times)
 6 -> 2: 50% (4/8 times)

--- Permutation test ---
Is the transition matrix more concentrated than random orderings produce?
King Wen matrix concentration: 2.3303
Random orderings equally or more concentrated: 2175/5000 (43.5%)
```

Gray Code Comparison

```
---
Gray code comparison
A Gray code is an ordering of binary values where each consecutive pair differs
by exactly 1 bit. For 6-bit hexagrams, a Gray code path would change exactly 1
line at every step — the smoothest possible sequence. The King Wen sequence does
NOT follow a Gray code (most transitions change 2-6 lines). This comparison
shows how much 'rougher' King Wen is than the theoretical minimum, suggesting
the ordering prioritizes something other than smooth transitions.
---
Metric                    Gray Code  King Wen
------------------------------------ ---------- ----------
Total path length                63     211
Mean change per transition          1.00    3.35
Max single transition              1      6
Min single transition              1      1
King Wen / Gray Code ratio               3.35x

Difference distribution:        Gray Code  King Wen
 1 line changes                 63      2
 2 line changes                 0      20
 3 line changes                 0      13
 4 line changes                 0      19
 6 line changes                 0       9
```

```
Gray code wave: ____________________________________________________________________________________
King Wen wave: ▙▄▟▄▗▄▟▗▄▟▗▄▟▄▗▄▟▗▄▟▗▄▟▄▗▄▟▗▄▟▗
```

```
6-line transitions          9       1       0

Waves:
 King Wen
 Fu Xi (binary)
 Mawangdui

--- What makes King Wen unique ---
Zero 5-line transitions: King Wen=0, Fu Xi=2, Mawangdui=0
Zero 0-line transitions: King Wen=0, Fu Xi=0, Mawangdui=0
```

Windowed Entropy

```
---
Windowed entropy analysis
Instead of one entropy value for the whole wave, we slide a window across it
and compute entropy locally. This reveals WHERE in the sequence structure
concentrates (low entropy = predictable) vs. dissolves (high entropy = varied).
Dips in the curve mark regions with repetitive transition patterns.
---
Window size: 15
Center Entropy  Visualization
------ -------  -------------
   8  1.7056  ########################
   9  1.5628  ######################
  10  1.7056  ########################
  11  1.8323  ##########################
  12  1.8323  ##########################
  13  1.8892  ##########################
  14  1.7968  ########################
  15  1.7968  ########################
  16  1.7232  ########################
  17  1.8295  ##########################
  18  1.8295  ##########################
  19  1.6729  #######################
  20  1.8295  ##########################
  21  1.8295  ##########################
  22  1.8295  ##########################
  23  1.8892  ##########################
  24  1.8892  ##########################
  25  1.8323  ##########################
  26  1.8892  ##########################
  27  1.8892  ##########################
  28  1.8892  ##########################
  29  1.8892  ##########################
  30  1.9086  ###########################
  31  1.9656  ############################
  32  1.8892  ##########################
  33  1.9656  ############################
  34  1.9656  ############################
  35  1.8892  ##########################
  36  1.8892  ##########################
  37  1.8062  ########################
  38  1.7465  ######################
  39  1.7465  ######################
  40  1.7465  ######################
  41  1.8062  ########################
  42  1.8062  ########################
  43  1.8062  ########################
  44  1.7465  ######################
  45  2.0226  ############################
```

```
  46  2.0226  ############################
  47  2.0662  #############################
  48  2.0419  #############################
  49  2.0662  #############################
  50  2.0662  #############################
  51  2.0419  #############################
  52  2.0662  #############################
  53  2.1736  ##############################
  54  2.2566  ################################
  55  2.2566  ################################
  56  2.2892  ################################

Mean windowed entropy: 1.8956
Min entropy: 1.5628 at position 9 (most structured)
Max entropy: 2.2892 at position 56 (most varied)

Entropy spark: ▁▁▁▁▁▅▁▁▁▁▁▁▁▁▅▅▁▅▅▅▅▅▅▅▅▅▅▁▁▁▁▁▁▁▁▅▅▅▅▅▅▅▅▅▅▅▅
```

Mutual Information

```
---
Mutual information: upper vs. lower trigram transitions
When two consecutive hexagrams differ, do the upper and lower trigrams change
independently, or is knowing that the upper trigram changed informative about
whether the lower trigram also changed? Mutual information quantifies this:
 0 bits = completely independent (knowing one tells you nothing about the other)
 higher = correlated (they tend to change together or stay together)
---
Joint distribution of trigram changes:
                Lower unchanged  Lower changed
 Upper unchanged       0 (  0.0%)     4 (  6.3%)
 Upper changed         5 (  7.9%)    54 ( 85.7%)

Mutual information: 0.0078 bits
Normalized MI: 0.0230 (0=independent, 1=perfectly correlated)

Mean MI of random permutations: 0.0201 bits
King Wen percentile: 7.6%

--- Full 8-state trigram mutual information ---
Using actual trigram identities (8 possible values each) rather than
binary changed/unchanged, which preserves more information.
MI(upper, lower) across all 64 hexagrams: 0.000000 bits
(Expected: 0.0 — since all 64 combinations appear exactly once,
the trigrams are perfectly independent in the static set.)
```

Yin-Yang Balance

```
---
Yin-yang balance wave
Each hexagram has 6 lines that are either yang (solid, 1) or yin (broken, 0).
Note: since the King Wen sequence contains all 64 possible hexagrams exactly
once, the total yin-yang count is necessarily balanced (192 each). This is a
mathematical tautology, not a property of the ordering. However, the LOCAL
balance as you move through the sequence IS a property of the ordering — it
shows where yang-dominant or yin-dominant hexagrams cluster together.
---
Total yang lines: 192 / 384 (50.0%)
Total yin lines:  192 / 384 (50.0%)
```

```
Mean yang per hexagram: 3.00

Running yang average (window=8):
 Pos  Yang  RunAvg   Visualization
  1   6   6.00  [         |         #]
  2   0   3.00  [         #         ]
  3   2   2.67  [        # |         ]
  4   2   2.50  [        #  |         ]
  5   4   2.80  [        #|         ]
  6   4   3.00  [         #         ]
  7   1   2.71  [        #|         ]
  8   1   2.50  [        #  |         ]
  9   5   2.38  [        #  |         ]
 10   5   3.00  [         #         ]
 11   3   3.12  [         #         ]
 12   3   3.25  [         |#         ]
 13   5   3.38  [         | #         ]
 14   5   3.50  [         | #         ]
 15   1   3.50  [         | #         ]
 16   1   3.50  [         | #         ]
 17   3   3.25  [         |#         ]
 18   3   3.00  [         #         ]
 19   2   2.88  [         #|         ]
 20   2   2.75  [        #|         ]
 21   3   2.50  [        #  |         ]
 22   3   2.25  [        #  |         ]
 23   1   2.25  [        #  |         ]
 24   1   2.25  [        #  |         ]
 25   4   2.38  [        #  |         ]
 26   4   2.50  [        #  |         ]
 27   2   2.50  [        #  |         ]
 28   4   2.75  [        #|         ]
 29   2   2.62  [        # |         ]
 30   4   2.75  [        #|         ]
 31   3   3.00  [         #         ]
 32   3   3.25  [         |#         ]
 33   4   3.25  [         |#         ]
 34   4   3.25  [         |#         ]
 35   2   3.25  [         |#         ]
 36   2   3.00  [         #         ]
 37   4   3.25  [         |#         ]
 38   4   3.25  [         |#         ]
 39   2   3.12  [         #         ]
 40   2   3.00  [         #         ]
 41   3   2.88  [         #|         ]
 42   3   2.75  [        #|         ]
 43   5   3.12  [         #         ]
 44   5   3.50  [         | #         ]
 45   2   3.25  [         |#         ]
 46   2   3.00  [         #         ]
 47   3   3.12  [         #         ]
 48   3   3.25  [         |#         ]
 49   4   3.38  [         | #         ]
 50   4   3.50  [         | #         ]
 51   2   3.12  [         #         ]
 52   2   2.75  [        #|         ]
 53   3   2.88  [         #|         ]
 54   3   3.00  [         #         ]
 55   3   3.00  [         #         ]
 56   3   3.00  [         #         ]
 57   4   3.00  [         #         ]
 58   4   3.00  [         #         ]
 59   3   3.12  [         #         ]
 60   3   3.25  [         |#         ]
```

```
 61   4  3.38  [          | #         ]
 62   2  3.25  [          |#          ]
 63   3  3.25  [          |#          ]
 64   3  3.25  [          |#          ]

Yang count spark: ▅▁▂▃▅▁▂▃▅▁▁▁▁▁▂▃▅▁▂▃▅▁▂▃▅▁▁▁▁▂▃▅▁▂▃▅▁▁▁▁▂▃▅▁▂▃▅▁
```

Neighborhoods

```
---
Hexagram neighborhoods
For each hexagram, which others are 'nearby' (differ by only 1 or 2 lines)?
Dense neighborhoods mean a hexagram has many close relatives; sparse ones are
more isolated. This also shows whether King Wen places neighbors near each
other in the sequence or scatters them.
---
01 ䷀ The Creative          d=1: [09, 10, 13, 14, 43, 44] (6 neighbors, mean seq dist: 21.2)
02 ䷁ The Receptive         d=1: [07, 08, 15, 16, 23, 24] (6 neighbors, mean seq dist: 13.5)
03 ䷂ Difficulty at the Beginning  d=1: [08, 17, 24, 42, 60, 63] (6 neighbors, mean seq dist: 32.7)
04 ䷃ Youthful Folly        d=1: [07, 18, 23, 41, 59, 64] (6 neighbors, mean seq dist: 31.3)
05 ䷄ Waiting           d=1: [09, 11, 43, 48, 60, 63] (6 neighbors, mean seq dist: 34.0)
06 ䷅ Conflict           d=1: [10, 12, 44, 47, 59, 64] (6 neighbors, mean seq dist: 33.3)
07 ䷆ The Army           d=1: [02, 04, 19, 29, 40, 46] (6 neighbors, mean seq dist: 19.0)
08 ䷇ Holding Together       d=1: [02, 03, 20, 29, 39, 45] (6 neighbors, mean seq dist: 18.7)
09 ䷈ Small Taming         d=1: [01, 05, 26, 37, 57, 61] (6 neighbors, mean seq dist: 26.2)
10 ䷉ Treading           d=1: [01, 06, 25, 38, 58, 61] (6 neighbors, mean seq dist: 25.8)
11 ䷊ Peace            d=1: [05, 19, 26, 34, 36, 46] (6 neighbors, mean seq dist: 18.7)
12 ䷋ Standstill         d=1: [06, 20, 25, 33, 35, 45] (6 neighbors, mean seq dist: 17.3)
13 ䷌ Fellowship         d=1: [01, 25, 30, 33, 37, 49] (6 neighbors, mean seq dist: 20.2)
14 ䷍ Great Possession       d=1: [01, 26, 30, 34, 38, 50] (6 neighbors, mean seq dist: 20.2)
15 ䷎ Modesty           d=1: [02, 36, 39, 46, 52, 62] (6 neighbors, mean seq dist: 28.8)
16 ䷏ Enthusiasm         d=1: [02, 35, 40, 45, 51, 62] (6 neighbors, mean seq dist: 27.8)
17 ䷐ Following          d=1: [03, 25, 45, 49, 51, 58] (6 neighbors, mean seq dist: 26.2)
18 ䷑ Work on the Decayed      d=1: [04, 26, 46, 50, 52, 57] (6 neighbors, mean seq dist: 25.8)
19 ䷒ Approach           d=1: [07, 11, 24, 41, 54, 60] (6 neighbors, mean seq dist: 20.5)
20 ䷓ Contemplation        d=1: [08, 12, 23, 42, 53, 59] (6 neighbors, mean seq dist: 19.5)
21 ䷔ Biting Through        d=1: [25, 27, 30, 35, 38, 51] (6 neighbors, mean seq dist: 13.3)
22 ䷕ Grace            d=1: [26, 27, 30, 36, 37, 52] (6 neighbors, mean seq dist: 12.7)
23 ䷖ Splitting Apart       d=1: [02, 04, 20, 27, 35, 52] (6 neighbors, mean seq dist: 14.7)
24 ䷗ Return           d=1: [02, 03, 19, 27, 36, 51] (6 neighbors, mean seq dist: 15.0)
25 ䷘ Innocence          d=1: [10, 12, 13, 17, 21, 42] (6 neighbors, mean seq dist: 11.5)
26 ䷙ Great Taming         d=1: [09, 11, 14, 18, 22, 41] (6 neighbors, mean seq dist: 11.8)
27 ䷚ Nourishment         d=1: [21, 22, 23, 24, 41, 42] (6 neighbors, mean seq dist: 7.8)
28 ䷛ Great Preponderance      d=1: [31, 32, 43, 44, 47, 48] (6 neighbors, mean seq dist: 12.8)
29 ䷜ The Abysmal         d=1: [07, 08, 47, 48, 59, 60] (6 neighbors, mean seq dist: 23.5)
30 ䷝ The Clinging         d=1: [13, 14, 21, 22, 55, 56] (6 neighbors, mean seq dist: 16.8)
31 ䷞ Influence          d=1: [28, 33, 39, 45, 49, 62] (6 neighbors, mean seq dist: 12.7)
32 ䷟ Duration           d=1: [28, 34, 40, 46, 50, 62] (6 neighbors, mean seq dist: 12.7)
33 ䷠ Retreat           d=1: [12, 13, 31, 44, 53, 56] (6 neighbors, mean seq dist: 16.2)
34 ䷡ Great Power         d=1: [11, 14, 32, 43, 54, 55] (6 neighbors, mean seq dist: 15.8)
35 ䷢ Progress           d=1: [12, 16, 21, 23, 56, 64] (6 neighbors, mean seq dist: 19.7)
36 ䷣ Darkening of the Light    d=1: [11, 15, 22, 24, 55, 63] (6 neighbors, mean seq dist: 19.7)
37 ䷤ The Family          d=1: [09, 13, 22, 42, 53, 63] (6 neighbors, mean seq dist: 19.0)
38 ䷥ Opposition         d=1: [10, 14, 21, 41, 54, 64] (6 neighbors, mean seq dist: 19.0)
39 ䷦ Obstruction         d=1: [08, 15, 31, 48, 53, 63] (6 neighbors, mean seq dist: 18.3)
40 ䷧ Deliverance         d=1: [07, 16, 32, 47, 54, 64] (6 neighbors, mean seq dist: 18.3)
41 ䷨ Decrease           d=1: [04, 19, 26, 27, 38, 61] (6 neighbors, mean seq dist: 18.5)
42 ䷩ Increase           d=1: [03, 20, 25, 27, 37, 61] (6 neighbors, mean seq dist: 19.5)
```

```
43 ䷐ Breakthrough          d=1: [01, 05, 28, 34, 49, 58] (6 neighbors, mean seq dist: 20.8)
44 ䷫ Coming to Meet        d=1: [01, 06, 28, 33, 50, 57] (6 neighbors, mean seq dist: 21.2)
45 ䷬ Gathering Together      d=1: [08, 12, 16, 17, 31, 47] (6 neighbors, mean seq dist: 23.8)
46 ䷭ Pushing Upward         d=1: [07, 11, 15, 18, 32, 48] (6 neighbors, mean seq dist: 24.8)
47 ䷮ Oppression           d=1: [06, 28, 29, 40, 45, 58] (6 neighbors, mean seq dist: 16.3)
48 ䷯ The Well            d=1: [05, 28, 29, 39, 46, 57] (6 neighbors, mean seq dist: 17.0)
49 ䷰ Revolution           d=1: [13, 17, 31, 43, 55, 63] (6 neighbors, mean seq dist: 18.7)
50 ䷱ The Cauldron          d=1: [14, 18, 32, 44, 56, 64] (6 neighbors, mean seq dist: 18.7)
51 ䷲ The Arousing          d=1: [16, 17, 21, 24, 54, 55] (6 neighbors, mean seq dist: 22.2)
52 ䷳ Keeping Still          d=1: [15, 18, 22, 23, 53, 56] (6 neighbors, mean seq dist: 22.5)
53 ䷴ Development           d=1: [20, 33, 37, 39, 52, 57] (6 neighbors, mean seq dist: 14.7)
54 ䷵ The Marrying Maiden      d=1: [19, 34, 38, 40, 51, 58] (6 neighbors, mean seq dist: 15.3)
55 ䷶ Abundance            d=1: [30, 34, 36, 49, 51, 62] (6 neighbors, mean seq dist: 13.7)
56 ䷷ The Wanderer          d=1: [30, 33, 35, 50, 52, 62] (6 neighbors, mean seq dist: 14.3)
57 ䷸ The Gentle           d=1: [09, 18, 44, 48, 53, 59] (6 neighbors, mean seq dist: 19.2)
58 ䷹ The Joyous           d=1: [10, 17, 43, 47, 54, 60] (6 neighbors, mean seq dist: 20.2)
59 ䷺ Dispersion           d=1: [04, 06, 20, 29, 57, 61] (6 neighbors, mean seq dist: 30.2)
60 ䷻ Limitation           d=1: [03, 05, 19, 29, 58, 61] (6 neighbors, mean seq dist: 31.2)
61 ䷼ Inner Truth          d=1: [09, 10, 41, 42, 59, 60] (6 neighbors, mean seq dist: 24.2)
62 ䷽ Small Preponderance      d=1: [15, 16, 31, 32, 55, 56] (6 neighbors, mean seq dist: 27.8)
63 ䷾ After Completion       d=1: [03, 05, 36, 37, 39, 49] (6 neighbors, mean seq dist: 34.8)
64 ䷿ Before Completion       d=1: [04, 06, 35, 38, 40, 50] (6 neighbors, mean seq dist: 35.2)

--- Neighborhood clustering in King Wen sequence ---
Mean sequence distance between Hamming-1 neighbors: 20.6
Mean sequence distance between any two hexagrams:  21.7
Hamming-1 neighbors are closer than average in the sequence (clustered).
```

Recurrence Plot

```
---
Recurrence plot
A recurrence plot shows where the difference wave repeats itself. Each cell (i,j)
is marked if the difference at position i equals the difference at position j.
Diagonal lines indicate repeated sequences; vertical/horizontal lines indicate
values that persist. Clusters of marks reveal recurring local patterns.
---
       1     2     3     4     5     6
   123456789012345678901234567890123456789012345678901234567890123
  1 +.........#.....#.........#.#........#..............#.......#.#
  2 .+....#.#..##.##....###..#.#..#....#....#.#...#......##...#....
  3 ..+##..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  4 ..#+#..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  5 ..##+..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  6 .....+...........#.#...#.....#.#.......#...#.#.#.......#.#...#.
  7 .#....+.#..##.##....###..#.#..#....#....#.#...#......##...#....
  8 ..###..+.#...#....#.....#.......###.#.#..#..#...###.....#......
  9 .#....#.+..##.##....###..#.#..#....#....#.#...#......##...#....
 10 ..###..#.+...#....#.....#.......###.#.#..#..#...###.....#......
 11 #.........+.....#.........#.#........#..............#.......#.#
 12 .#....#.#..+#.##....###..#.#..#....#....#.#...#......##...#....
 13 .#....#.#..#+.##....###..#.#..#....#....#.#...#......##...#....
 14 ..###..#.#...+....#.....#.......###.#.#..#..#...###.....#......
 15 .#....#.#..##.+#....###..#.#..#....#....#.#...#......##...#....
 16 .#....#.#..##.#+....###..#.#..#....#....#.#...#......##...#....
 17 #.........#.....+.........#.#........#..............#.......#.#
 18 .....#...........+.#...#.....#.#.......#...#.#.#.......#.#...#.
 19 ..###..#.#...#....+.....#.......###.#.#..#..#...###.....#......
 20 .....#...........#.+...#.....#.#.......#...#.#.#.......#.#...#.
 21 .#....#.#..##.##....+##..#.#..#....#....#.#...#......##...#....
 22 .#....#.#..##.##....#+#..#.#..#....#....#.#...#......##...#....
```

```
 23 .#....#.#..##.##....##+..#.#..#....#....#.#...#......##...#....
 24 .....#...........#.#...+.....#.#.......#...#.#.#.......#.#...#.
 25 ..###..#.#...#....#.....+.......###.#.#..#..#...###.....#......
 26 .#....#.#..##.##....###..+.#..#....#....#.#...#......##...#....
 27 #.........#.....#.........+.#........#..............#.......#.#
 28 .#....#.#..##.##....###..#.+..#....#....#.#...#......##...#....
 29 #.........#.....#.........#.+........#..............#.......#.#
 30 .....#...........#.#...#.....+.#.......#...#.#.#.......#.#...#.
 31 .#....#.#..##.##....###..#.#..+....#....#.#...#......##...#....
 32 .....#...........#.#...#.....#.+.......#...#.#.#.......#.#...#.
 33 ..###..#.#...#....#.....#.......+##.#.#..#..#...###.....#......
 34 ..###..#.#...#....#.....#.......#+#.#.#..#..#...###.....#......
 35 ..###..#.#...#....#.....#.......##+.#.#..#..#...###.....#......
 36 .#....#.#..##.##....###..#.#..#....+....#.#...#......##...#....
 37 ..###..#.#...#....#.....#.......###.+.#..#..#...###.....#......
 38 #.........#.....#.........#.#........+..............#.......#.#
 39 ..###..#.#...#....#.....#.......###.#.+..#..#...###.....#......
 40 .....#...........#.#...#.....#.#.......+...#.#.#.......#.#...#.
 41 .#....#.#..##.##....###..#.#..#....#....+.#...#......##...#....
 42 ..###..#.#...#....#.....#.......###.#.#..+..#...###.....#......
 43 .#....#.#..##.##....###..#.#..#....#....#.+...#......##...#....
 44 .....#...........#.#...#.....#.#.......#...+.#.#.......#.#...#.
 45 ..###..#.#...#....#.....#.......###.#.#..#..+...###.....#......
 46 .....#...........#.#...#.....#.#.......#...#.+.#.......#.#...#.
 47 .#....#.#..##.##....###..#.#..#....#....#.#...+......##...#....
 48 .....#...........#.#...#.....#.#.......#...#.#.+.......#.#...#.
 49 ..###..#.#...#....#.....#.......###.#.#..#..#...+##.....#......
 50 ..###..#.#...#....#.....#.......###.#.#..#..#...#+#.....#......
 51 ..###..#.#...#....#.....#.......###.#.#..#..#...##+.....#......
 52 ..........................................................+.......#...
 53 #.........#.....#.........#.#........#..............+.......#.#
 54 .#....#.#..##.##....###..#.#..#....#....#.#...#......+#...#....
 55 .#....#.#..##.##....###..#.#..#....#....#.#...#......#+...#....
 56 .....#...........#.#...#.....#.#.......#...#.#.#.......+.#...#.
 57 ..###..#.#...#....#.....#.......###.#.#..#..#...###.....+......
 58 .....#...........#.#...#.....#.#.......#...#.#.#.......#.+...#.
 59 .#....#.#..##.##....###..#.#..#....#....#.#...#......##...+....
 60 ..........................................................#.......+...
 61 #.........#.....#.........#.#........#..............#.......+.#
 62 .....#...........#.#...#.....#.#.......#...#.#.#.......#.#...+.
 63 #.........#.....#.........#.#........#..............#.......#.+

Recurrence rate: 952/3906 (24.4%)
```

DNA Codons

```
---
DNA codon mapping
Both the I Ching (64 hexagrams) and the genetic code (64 codons) are systems
of 64 elements built from binary-like pairs: yin/yang for hexagrams, and the
base pairs A-T and G-C for DNA. Each codon is a sequence of 3 bases from the
set {A, C, G, U}, giving 4^3 = 64 possibilities. While this is a numerical
coincidence (both are 2^6 = 64), comparing their structural properties is
instructive about combinatorial systems in general.
---
Pos | Hex | Binary | Codon | Amino Acid | Name
----|-----|--------|-------|------------|----
01 | ䷀ | 111111 | GGG | Gly    | The Creative
02 | ䷁ | 000000 | UUU | Phe    | The Receptive
03 | ䷂ | 010001 | CUC | Leu    | Difficulty at the Beginning
04 | ䷃ | 100010 | AUA | Ile    | Youthful Folly
```

05	䷓	010111	CCG	Pro	Waiting
06	䷅	111010	GAA	Glu	Conflict
07	䷆	000010	UUA	Leu	The Army
08	䷇	010000	CUU	Leu	Holding Together
09	䷈	110111	GCG	Ala	Small Taming
10	䷉	111011	GAG	Glu	Treading
11	䷊	000111	UCG	Ser	Peace
12	䷋	111000	GAU	Asp	Standstill
13	䷌	111101	GGC	Gly	Fellowship
14	䷍	101111	AGG	Arg	Great Possession
15	䷎	000100	UCU	Ser	Modesty
16	䷏	001000	UAU	Tyr	Enthusiasm
17	䷐	011001	CAC	His	Following
18	䷑	100110	ACA	Thr	Work on the Decayed
19	䷒	000011	UUG	Leu	Approach
20	䷓	110000	GUU	Val	Contemplation
21	䷔	101001	AAC	Asn	Biting Through
22	䷕	100101	ACC	Thr	Grace
23	䷖	100000	AUU	Ile	Splitting Apart
24	䷗	000001	UUC	Phe	Return
25	䷘	111001	GAC	Asp	Innocence
26	䷙	100111	ACG	Thr	Great Taming
27	䷚	100001	AUC	Ile	Nourishment
28	䷛	011110	CGA	Arg	Great Preponderance
29	䷜	010010	CUA	Leu	The Abysmal
30	䷝	101101	AGC	Ser	The Clinging
31	䷞	011100	CGU	Arg	Influence
32	䷟	001110	UGA	Stop	Duration
33	䷠	111100	GGU	Gly	Retreat
34	䷡	001111	UGG	Trp	Great Power
35	䷢	101000	AAU	Asn	Progress
36	䷣	000101	UCC	Ser	Darkening of the Light
37	䷤	110101	GCC	Ala	The Family
38	䷥	101011	AAG	Lys	Opposition
39	䷦	010100	CCU	Pro	Obstruction
40	䷧	001010	UAA	Stop	Deliverance
41	䷨	100011	AUG	Met	Decrease
42	䷩	110001	GUC	Val	Increase
43	䷪	011111	CGG	Arg	Breakthrough
44	䷫	111110	GGA	Gly	Coming to Meet
45	䷬	011000	CAU	His	Gathering Together
46	䷭	000110	UCA	Ser	Pushing Upward
47	䷮	011010	CAA	Gln	Oppression
48	䷯	010110	CCA	Pro	The Well
49	䷰	011101	CGC	Arg	Revolution
50	䷱	101110	AGA	Arg	The Cauldron
51	䷲	001001	UAC	Tyr	The Arousing
52	䷳	100100	ACU	Thr	Keeping Still
53	䷴	110100	GCU	Ala	Development
54	䷵	001011	UAG	Stop	The Marrying Maiden
55	䷶	001101	UGC	Cys	Abundance
56	䷷	101100	AGU	Ser	The Wanderer
57	䷸	110110	GCA	Ala	The Gentle
58	䷹	011011	CAG	Gln	The Joyous
59	䷺	110010	GUA	Val	Dispersion
60	䷻	010011	CUG	Leu	Limitation
61	䷼	110011	GUG	Val	Inner Truth
62	䷽	001100	UGU	Cys	Small Preponderance
63	䷾	010101	CCC	Pro	After Completion
64	䷿	101010	AAA	Lys	Before Completion

Unique amino acids in King Wen order: 21
Total possible amino acids + stop: 21

```
--- Degeneracy comparison ---
In DNA, many single-base changes preserve the amino acid (the code is 'degenerate').
Do single-line hexagram changes preserve the mapped amino acid?
Single-line changes that preserve amino acid: 100/384 (26.0%)
```

Shannon Entropy

```
---
Shannon entropy analysis
Entropy measures disorder: high entropy means the difference values are spread
evenly across all possibilities (random-looking), low entropy means certain values
dominate (structured). We compare the King Wen wave's entropy against thousands
of random permutations of the same 64 hexagrams. If King Wen's entropy is
unusually low, the sequence is more structured than random chance would produce.
---
King Wen difference wave entropy: 2.0759 bits
Maximum possible entropy (uniform): 2.8074 bits
Mean entropy of random permutations: 2.1935 bits
Min random entropy observed: 1.7506 bits
Max random entropy observed: 2.4967 bits
King Wen percentile: 12.6% (lower = more structured)

--- Distribution comparison ---
Value   King Wen Expected (random avg)
 0       0        0.0
 1       2        6.0
 2      20        14.9
 3      13        20.2
 4      19        14.9
 5       0        6.0
 6       9        0.9
```

Path Analysis

```
---
Path analysis (graph theory)
Imagine the 64 hexagrams as cities on a map, where the 'distance' between any
two is the number of lines that differ. The King Wen sequence is a route that
visits all 64 cities. Is it a short, efficient route (nearby hexagrams placed
next to each other), or a long, wandering one? We compare its total distance
against random routes and a greedy 'always go to the nearest unvisited city'
algorithm. This reveals whether the King Wen ordering was optimized for smooth
transitions or had other priorities.
---
King Wen total path length:  211 (sum of all line changes)
Greedy nearest-neighbor:   75
Mean random path length:   192.0
Min random observed:     159
Max random observed:     227
King Wen percentile:     97.7% (lower = shorter path)

--- Theoretical bounds ---
Minimum possible (63 transitions of 1): 63
Maximum possible (63 transitions of 6): 378
King Wen uses 55.8% of maximum path length

--- Pair-constrained comparison ---
The above compares against fully random orderings. A fairer comparison is
against random orderings that also preserve the pair structure (each pair
```

```
of hexagrams stays adjacent). This is the right null model for asking
whether King Wen's path length is unusual GIVEN its pair constraint.
Mean pair-constrained path length: 214.1
Min pair-constrained observed:   191
Max pair-constrained observed:   242
King Wen percentile (pair-constrained): 29.3%
```

Constraint Satisfaction

```
---
Constraint satisfaction analysis
The King Wen sequence satisfies several constraints simultaneously:
 1. All 32 consecutive pairs are either reverse or inverse (never unrelated)
 2. No consecutive hexagrams differ by exactly 5 lines
 3. No consecutive hexagrams are identical (0-line transitions)
How rare is it for a random ordering to satisfy all these at once? We test
random permutations against each constraint individually and combined.
---
Results from 10,000 random permutations:
 All pairs reverse/inverse:    0 (0.000%)
 No 5-line transitions:      13 (0.13%)
 Both constraints together:    0 (0.0000%)
 No random permutation satisfied both constraints.
 Statistical note: 0/10,000 gives a 95% upper bound of <0.0300%
 (i.e., the true rate is less than ~1 in 3,333, but we cannot
 distinguish 'impossible' from 'extremely rare' with this sample size).
```

Bootstrap

```
---
Bootstrap confidence intervals
The Monte Carlo results (e.g., '0.18% of random orderings avoid 5-line
transitions') are estimates based on sampling. Bootstrap resampling quantifies
how precise those estimates are by repeatedly resampling from the results and
computing confidence intervals. Narrower intervals = more reliable estimates.
---
Base trials: 10,000
Bootstrap resamples: 1000

No-5-line-transition rate: 0.070%
95% confidence interval: [0.020%, 0.130%]
Interval width: 0.110 percentage points

Approximately 1 in 1429 random orderings
95% CI: 1 in 769 to 1 in 5000
```

Monte Carlo

```
---
Monte Carlo analysis (10,000 random permutations)
The King Wen sequence has a striking property: no two consecutive hexagrams
differ by exactly 5 lines. With 6 lines per hexagram and 7 possible difference
values (0-6), is avoiding 5 remarkable or just a coincidence? To find out, we
randomly shuffle the 64 hexagrams thousands of times and check how often a
random ordering also avoids 5-line transitions. The rarer it is, the more
likely the King Wen sequence was intentionally designed with this constraint.
```

```
---
Permutations with no 5-line transitions: 14/10,000 (0.14%)
Approximately 1 in 714 random orderings share this property.
```